SOMAIYA VIDYAVIHAR UNIVERSITY

27/05/2025(E)

Semester: January 2025 –April 2025		
Maximum Marks: 100 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01 Programme: B Tech Computer Engg	Class: SY	Semester: III(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 116U01C303	Name of the Course: Computer Organization and Architecture	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	What is IAS Computer? State the importance of Von Neumann Model?	5
ii)	Define two's complement and hence support your answer with an example	5
iii)	What is meant by CPU architecture?	5
iv)	Draw the memory hierarchy.	5
v)	What is DMA?	5
vi)	Define a bus and the types of buses i.e dedicated and multiplexed buses.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Draw the formats of IEEE-754 floating point representation i.e Single Precision (32 bits) and Double Precision (64 bits).	5
ii)	Differentiate between Restoring and Non Restoring Division. Also state the role of accumulator in both the approaches.	5
	OR	
Q2 A	Solve by using Booths Multiplication algorithm for M=10 and Q=15 , use n=5 bits	10
Q 2 B	Solve any **One**	**10**
i)	Solve the following using Booth's recoding algorithm for M=5 and Q=5. Show all steps in detail.	10
ii)	Draw the flowcharts for floating point division and multiplication and justify the importance of Bias field in the calculations.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain the register organization of 8086	10
ii)	Describe the instruction cycle state diagram.	10
iii)	Differentiate between RISC and CISC processor	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain the characteristics of memory in detail	10
ii)	What is RAM , hence differentiate between SRAM and DRAM.	10
iii)	Analyse the RAID Levels 0-6 and hence comment on the applications of each level.	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Define PCI Bus. What is difference between SCSI and PCI	5
ii)	Differentiate between fixed and floating points number. Support your answer with a suitable example.	5
iii)	What is the importance of Flag register.	5
iv)	Describe the applications of Virtual Memory.	5
v)	State the Flynn's classification in brief	5
vi)	Mention the importance of six stage pipeline architecture	5